Fundamental Calculus/Differential and Integral Calculus

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Reference : Calculus 10-th Edition by Howard Anton, Irl Bivens and Stephen Davis

Definition of Constant

- A constant can simply be defined as a value used in mathematics for algebraic expressions and equations. A constant does not change over time and has a fixed value. For example, the size of a shoe or cloth or any apparel will not change at any point of time by itself.
- In an algebraic expression, $x + y = 8$, 8 is a constant value and it cannot be changed.

Definition of Variables

- Variables are the terms which can change or vary over time. It does not remain constant, unlike constant. For example, the height and weight of a person do not remain constant always and hence they are variables.
- In an algebraic expression, $x + y = 8$, x and y are the variables and can be changed.

Inequality

- An inequality compares two values, showing if one is less than, greater than, or simply not equal to another value.
- **$a \neq b$** says that a is not equal to b
- **$a < b$** says that a is less than b

- **$a > b$** says that a is greater than b
  (those two are known as strict inequality)
- **$a \leq b$** means that a is less than or equal to b
- **$a \geq b$** means that a is greater than or equal to b.

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Interval Notation

- Interval notation is a common way to express the solution set to an inequality, and it’s important because it’s how you express solution sets in calculus.

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Inequalities and Interval Notation

Intervals

Name of interval	Notation	Inequality description	Number line representation
Finite and closed	[a, b]	$a \le x \le b$	a b
Finite and open	(a, b)	$a < x < b$	a b
Finite and half-open	[a, b)	$a \le x < b$	
	(a, b]	$a < x \le b$	a b
Infinite and closed	$(-\infty, b]$	$-\infty < x \le b$	b
	$[a, +\infty)$	$a \le x < +\infty$	a
Infinite and open	$(-\infty, b)$	$-\infty < x < b$	b
	$(a, +\infty)$	$a < x < +\infty$	a
Infinite and open	$(-\infty, +\infty)$	$-\infty < x < +\infty$	

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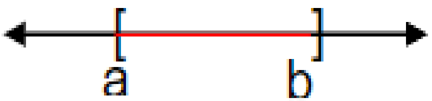
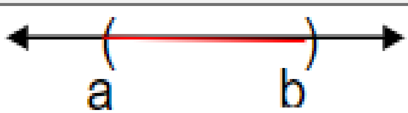
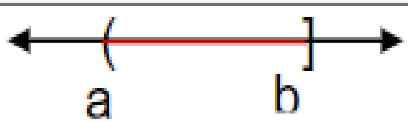
Interval Notation Review

Let a and b represent two real numbers with $a < b$.

Type of Interval	Interval Notation	Set-Builder Notation	Graph on the Number Line
Closed Interval	[a,b]	$\{x \mid a \leq x \leq b\}$	
Open Interval	(a,b)	$\{x \mid a < x < b\}$	
Half-Open Interval	(a,b]	$\{x \mid a < x \leq b\}$	
Half-Open Interval	[a,b)	$\{x \mid a \leq x < b\}$	

Chapter:0.1(page-25)

- Function
- Domain and Range of a function.

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Function

- A **function is** a **relationship between** quantities where there **is** one output for every input. If you have more than one output for a particular input, then the quantities represent a **relation.**

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DEFINITION OF A FUNCTION

- Many scientific laws and engineering principles describe how one quantity depends on
another. This idea was formalized in 1673 by Gottfried Wilhelm Leibniz (see p. xx) who
coined the term *function* to indicate the dependence of one quantity on another, as described
in the following definition.
-
0.1.1 definition If a variable *y (Grade)* depends on a variable *x(writing)* in such a way that each
value of *x* determines exactly one value of *y*, then we say that ***y** (Grade) is a function of **x** (writing)***.

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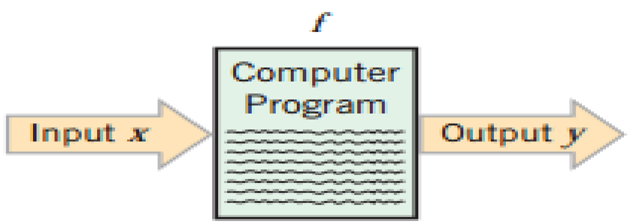
DEFINITION OF A FUNCTION

0.1.2 definition A ***function f*** **is a rule that associates a unique output with each input. If the input is denoted by *x*, then the output is denoted by *f (x)* (read "*f* of *x*") that is** $y = f(x)$.



Thus, the function *f* (the computer program) associates a unique output *y* with each input *x.* In this definition the term *unique* means "exactly one." Thus, a function cannot assign two different outputs to the same input.

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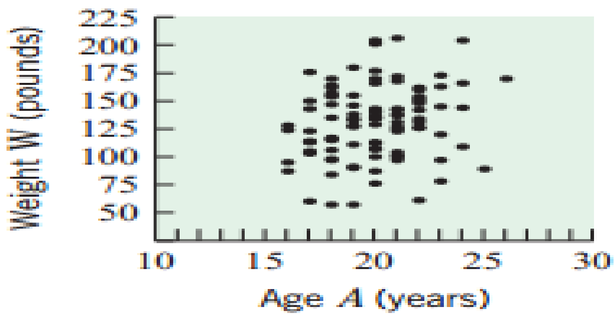
Calculus

- Calculus is **the study of how things change**. It provides a framework for modeling systems in which there is change, and a way to deduce the predictions of such models.
- we actually use calculus quite often in our daily lives. **Various fields such as engineering, medicine, biological research, economics, architecture, space science, electronics, statistics, and pharmacology** all benefit from the use of calculus.
- The notion of rate of change is so important.
  What is the rate of change of prices ? What is the rate at which temperature of a body changes in heat conduction ? What is the rate of change of velocity of an object ? What is the rate of increase of the population under certain conditions ?



This plot does *not* describe *W* as a function of *A* because there are some values of *A* with more than one corresponding value of *W*. This is to be expected, since two people with the same age can have different weights.

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Real Life Application

- If we consider the grading policy or examination system of any course, different students perform differently according to their efforts and knowledge. Here the grading policy or the examination system are the function of performance.

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Domain and Range of function

- Example: $y(Grade) = f(x) = f(performance)$.

- If x and y are related by the equation $y = f(x)$,

then the set of all allowable inputs (*x*-values) is called the *domain* of *f*, and the set of outputs (*y*-values) that result when *x* varies over the domain is called the *range* of *f*.

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Grade

Question Marks	Obtaining Marks
1. 10	10
2.10	8
3.10	6

Assesment
1→10
2→8
3→6

Individual output for Individual input	
(1,10)	
(2,8)	
(3,6)	
Total output=24	

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Consider the function $y = f(x) = x$

Input x	Assesment y	Individual output y for Individual input x
1. $\boldsymbol{x} = -2,$	-2→-2	(-2,-2)
2. $\boldsymbol{x} = -1$	-1→-1	(-1,-1)
3. $x = 0$	0→0	(0,0)
4. $x = 1$	1→1	(1,1)
5. $x = 2$	2→2	(2,2)
6. $x = 3$	3→3	(3,3)

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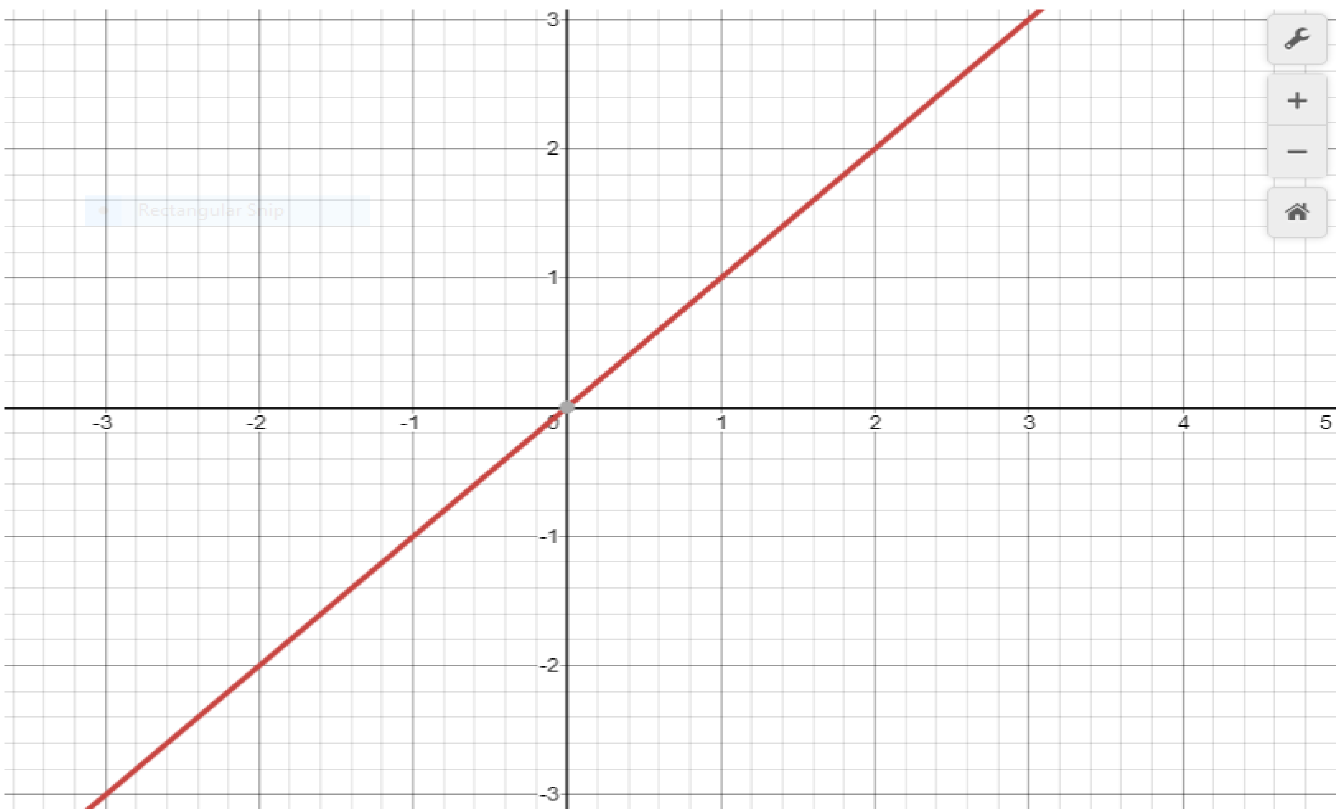
Output y(Range)



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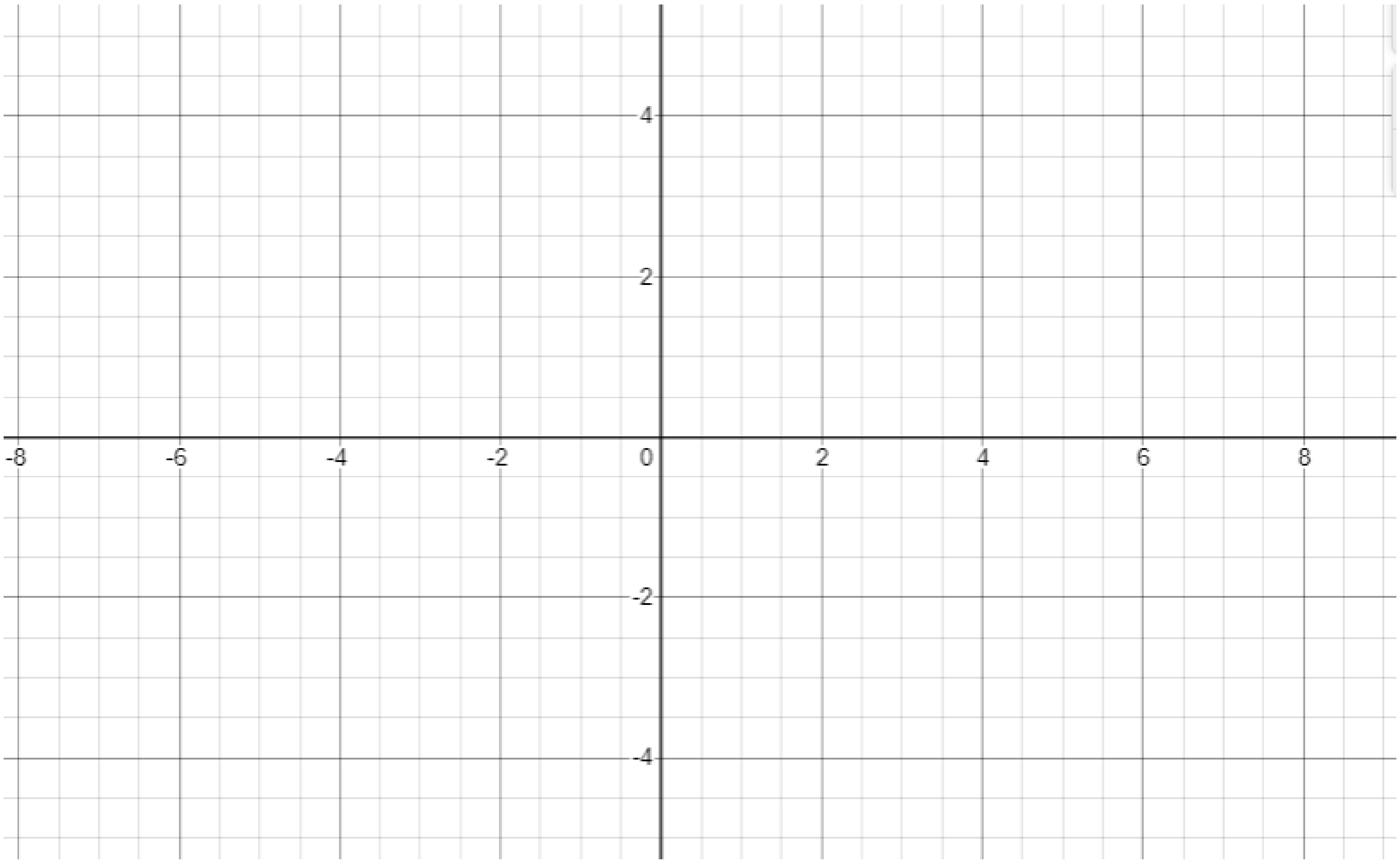
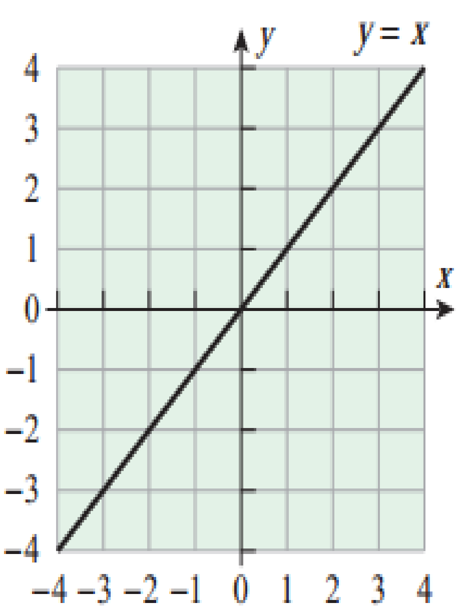
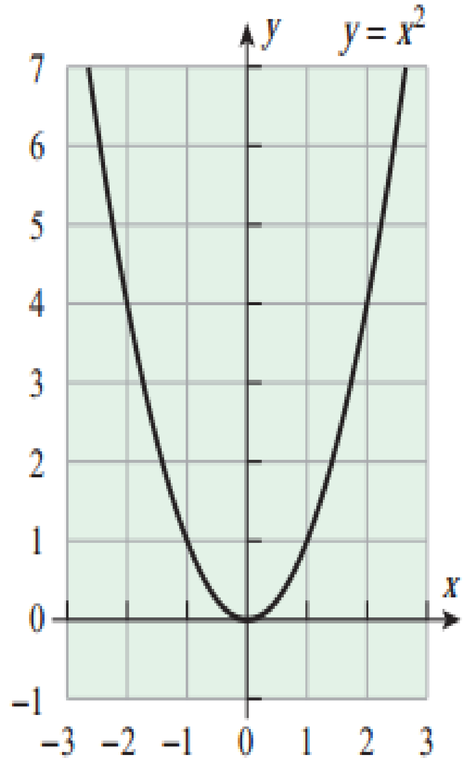
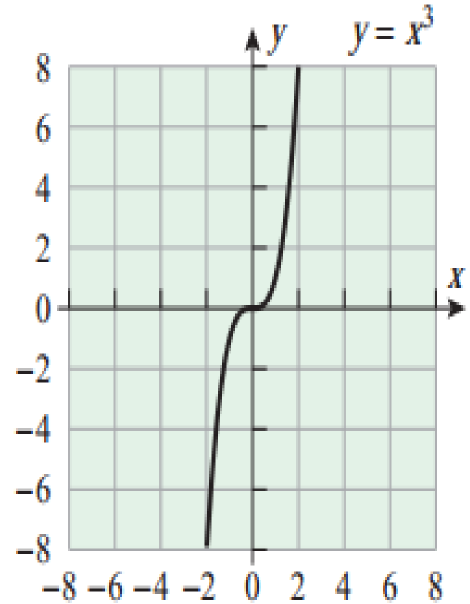


4
2
-8
-6
-4
-2
0
2
4
6
8
-2
-4

Graph of Functions







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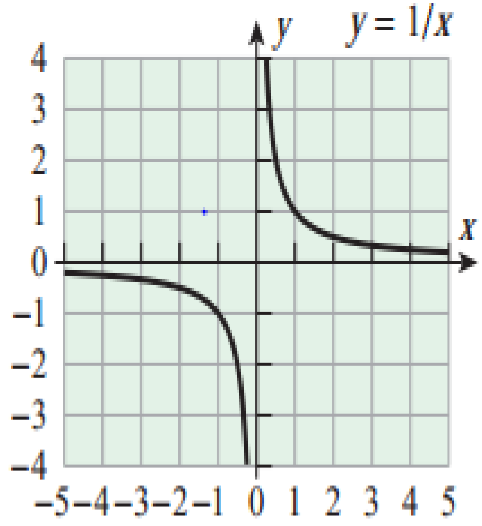
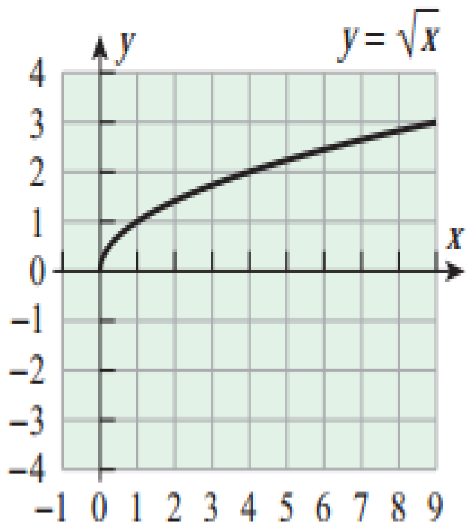
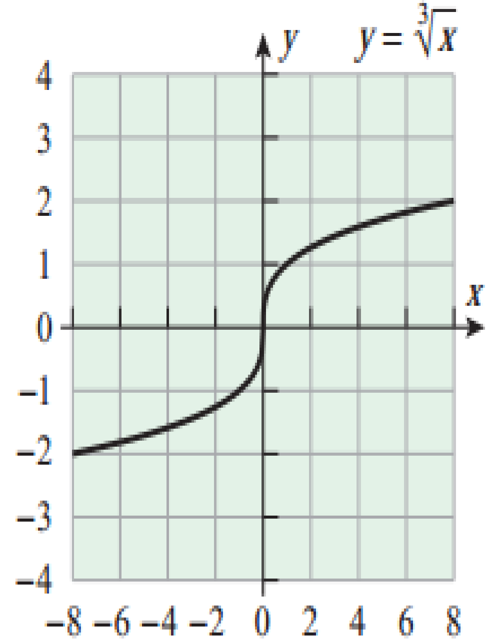
Graph of Functions







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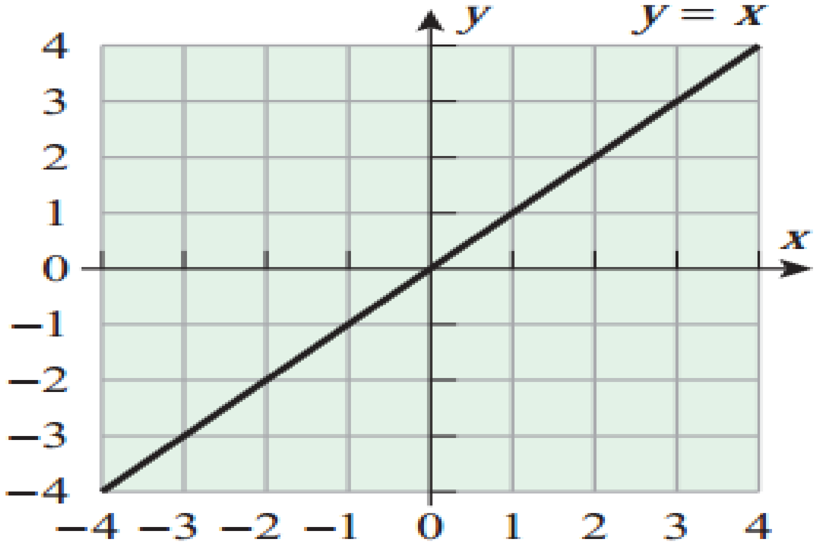
Domain and range of $y = f(x) = x, Domain\ is$ $(-\infty, +\infty)$, and Range is $(-\infty, +\infty)$



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Exercise(Homework)

1. Sketch the graph of $y = P(x)$, also find their domain and range.

$(a)\ y = x^2$

$(b)\ y = x^3$

$(c)\ y = x^{\frac{1}{3}} = \sqrt[3]{x}$

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